The documentation and process conversion measures necessary to comply with this revision shall be completed by 28 September 2015.

INCH–POUND

MIL–PRF–19500/207D
27 June 2015
SUPERSEDING
MIL–S–9500/207C(EL)
9 December 1966

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, SWITCHING, MEDIUM–POWER, THROUGH-HOLE MOUNT, TYPES 2N1479, 2N1480, 2N1481 AND 2N1482, QUALITY LEVEL JAN

Inactive for new design after 7 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–PRF–19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, NPN, medium power transistors for use in compatible equipment circuits. One level of product assurance (JAN) is provided for all encapsulated devices.

1.2 Package outlines. The device package outlines are as follows: TO–205AA (formerly modified TO–5) (without suffix S, see 1.5.3) or a TO–205AD (formerly modified TO–39) (with suffix S, see 1.5.3) in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1)	V_{CBO}	V_{CEO}	V_{CEX}	V_{EBO}	I_B	I_C	$R_{\theta JC}$	T_{STG}
	W	V dc	V dc	V dc	V dc	A dc	A dc	ºC/mW	°C
2N1479	1	60	40	60	12	1.0	1.5	35	
2N1480	1	100	55	100	12	1.0	1.5	35	–65 to
2N1481	1	60	40	60	12	1.0	1.5	35	+200
2N1482	1	100	55	100	12	1.0	1.5	35	

(1) This power dissipation is for 1000 hours expected life at T_A = 25ºC.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25ºC.

Limits	h_{FE} (1) V_{CE} = 4 V dc I_C = 200 mA dc		$V_{CE(sat)}$ (1) I_C = 200 mA dc (2)	V_{BE} (1) I_C = 200 mA dc V_{CE} = 4 V dc	I_{CBO} V_{CE} = 30 V dc	I_{EBO} V_{EB} = 12 V dc	f_{ab} I_C = 5 mA dc V_{CB} = 28 V dc
	2N1479 2N1480	2N1481 2N1482					
			V dc	V dc	µA dc	µA dc	kHz
Min	20	35					800
Max	60	100	0.75	1.5	5	10	

(1) Pulsed (see 4.5.1).
(2) For types 2N1479, 2N1480 I_B = 20 m A dc; for types 2N1481, 2N1482 I_B = 10 m A dc.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL–PRF–19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The only quality level designator for encapsulated devices that is applicable for this specification sheet is the base quality level "JAN" that uses no modifiers.

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "1479", "1480, "1481", and "1482".

1.5.3 Suffix symbols. The suffix symbol "S" is used on devices that have a shortened lead length: 0.5 inch (12.7 mm) minimum to .75 inch (19.1 mm) maximum. Devices with standard length leads (see figure 1) use no suffix.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS–19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL–PRF–19500 – Semiconductor Devices, General Specification for.

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DEPARTMENT OF DEFENSE STANDARDS

MIL–STD–750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL–PRF–19500 and as specified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL–PRF–19500 and as follows:

t_{on} - - - - - - - $(t_r + t_d)$
t_{off} - - - - - - - $(t_s + t_f)$

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL–PRF–19500 and herein. The device package style is a TO–205AA (formerly modified TO–5) or a TO–205AD (formerly modified TO–39) in accordance with figure 1 for all device types.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL–STD–750, MIL–PRF–19500, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2).

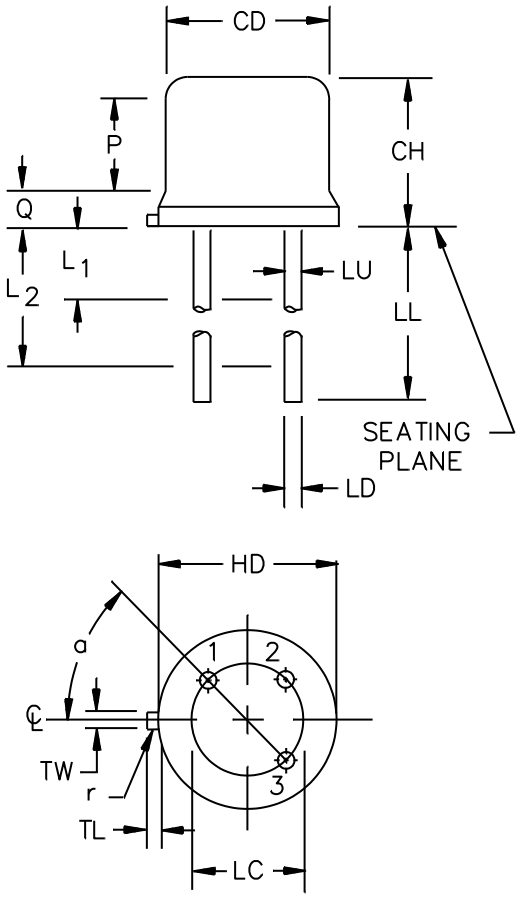
3.4.2 Pin-out. The pin-out of the device shall be as shown on figure 1. Terminal 1 is the emitter, terminal 2 is the base, and terminal 3 is the collector. The collector shall be electrically connected to the case.

3.4.3 Terminal-lead length (see 1.5.3 and figure 1). Terminal-lead length(s) other than that specified on figure 1 or by the PIN may be furnished under contract or order (see 6.2) when the devices covered herein are required directly for particular equipment-circuit installation or for automatic-assembly-technique programs. Where such longer or shorter leads are required and provided, it shall not be construed as adversely affecting the Qualified-product status of the device, or applicable JAN marking (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics shall be as specified in 1.3, 1.4, and table I herein.

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	3
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		4
LD	.016	.019	0.41	0.48	5, 6
LL	See notes 6, 7, and 8				
LU	.016	.019	0.41	0.48	5, 6
L_1		.050		1.27	5, 6
L_2	.250		6.35		5, 6
P	.100		2.54		9
Q		.030		0.76	3
TL	.029	.045	0.74	1.14	10, 11
TW	.028	.034	0.71	0.86	10
r		.010		0.25	12
α	45° TP		45° TP		4



NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Lead 1 = emitter, lead 2 = base, lead 3 = collector. The collector shall be internally connected to the case.
3. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
4. Leads at gauge plane .054 +.001 –.000 inch (1.37 +0.03 –0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauging procedure.
5. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in and beyond LL minimum.
6. All three leads.
7. For the modified TO–205AA (formerly TO–5) package (PINs without the S suffix), dimension LL is 1.500 inch (38.10 mm) minimum and 1.750 inch (44.45 mm) maximum.
8. For the modified TO–205AD (formerly TO–39) package (PINs with the S suffix), dimension LL is .500 inch (12.70 mm) minimum and .750 inch (19.05 mm) maximum.
9. Body contour optional within zone defined by dimensions HD, CD, and Q.
10. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 (0.28 mm).
11. Dimension TL measured from maximum HD.
12. Dimension r (radius) applies to both inside corners of tab.
13. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions and configuration of TO–205AA and TO–205AD package.

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3.6 Marking. Marking shall be in accordance with MIL–PRF–19500.

3.7 Workmanship. Devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL–PRF–19500, and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening. Screening is not applicable for devices compliant to this specification sheet.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL–PRF–19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL–PRF–19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VIB of MIL–PRF–19500 and herein.

Subgroup	Method	Conditions
B1	2026	Omit preconditioning.
B3	1027	T_A = 25ºC; P_T = 600 mW; V_{CE} = 30 V dc
B5	3131	See 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E–VII of MIL–PRF–19500 and as follows herein.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	T_A = 25ºC; P_T = 600 mW; V_{CE} = 30 V dc.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E–IX of MIL–PRF–19500 and as specified in table II herein.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL–STD–750.

4.5.2 Steady-state thermal impedance. The steady-state thermal impedance measurements shall be performed in accordance with test method 3131 of MIL–STD–750 using the guidelines in that test method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). The steady-state thermal impedance shall statistically set by the supplier and approved by the qualifying activity. See table II, subgroup 4 herein.

TABLE I. Group A inspection.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Steady-state thermal impedance 2/	3131	See 4.5.2	$R_{\theta JC}$		35	°C/mW
Collector to base, cutoff current	3036	Bias condition D				
2N1479, 2N1481		V_{CB} = 30 V dc.	I_{CBO1}		5	µA dc
2N1480, 2N1482		V_{CB} = 50 V dc.	I_{CBO1}		5	µA dc
Emitter to base, cutoff current	3061	Bias condition D, V_{EB} = 12 V dc.	I_{EBO}		10	µA dc
Collector to emitter breakdown voltage	3011	Bias condition D, I_E = 50 mA dc, pulsed (see 4.5.1)				V dc
2N1479, 2N1481		V_{CB} = 30 V dc.	$V_{(BR)CEO}$	40		V dc
2N1480, 2N1482		V_{CB} = 50 V dc.	$V_{(BR)CEO}$	50		V dc
Collector to emitter breakdown voltage	3011	Bias condition A, I_C = 0.25 mA dc, V_{EB} = 1.5 V dc; pulsed (see 4.5.1)				V dc
2N1479, 2N1481			$V_{(BR)CEX}$	60		V dc
2N1480, 2N1482			$V_{(BR)CEX}$	100		V dc
Forward-current transfer ratio	3076	I_C = 200 mA dc; V_{CE} = 4 V dc, pulsed (see 4.5.1)				
2N1479, 2N1480			h_{FE}	20	60	
2N1481, 2N1482			h_{FE}	35	100	
Collector to emitter voltage (saturated)	3071	I_C = 200 mA dc, pulsed (see 4.5.1).				V dc
2N1479, 2N1480		I_B = 20 mA dc	$V_{CE(sat)}$		0.75	V dc
2N1481, 2N1482		I_B = 10 mA dc	$V_{CE(sat)}$		0.75	V dc

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL–STD–750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – continued.						
Base to emitter voltage (unsaturated)	3066	Test condition B; I_C = 200 mA dc V_{CE} = 4 V dc; pulsed (see 4.5.1)	V_{BE}		1.5	V dc
Subgroup 3						
High temperature operation		T_A = +170°C				
Collector to base cut-off current	3036	Bias condition D				
2N1479, 2N1481		V_{CB} = 30 V dc	I_{CBO2}		0.75	µA dc
2N1480, 2N1482		V_{CB} = 50 V dc	I_{CBO2}		0.75	µA dc
Low temperature operation		T_A = –55°C				
Forward-current transfer ratio	3076	I_C = 200 mA dc, V_{CE} = 4 V dc,				
2N1479, 2N1480			h_{FE}	10		
2N1481, 2N1482			h_{FE}	17		
Subgroup 4						
Pulse response	3251	Test condition A, R_C = 59 ohms, V_{CC} = 12 V dc, $I_{B(0)} = I_{B(2)}$ = 8.5 mA dc, $I_{B(1)}$ = 20 mA dc	$t_{on} + t_{off}$		25	µsec
Small-signal, short-circuit, forward-current transfer ratio cut off frequency	3301	I_C = 5 mA dc, V_{CB} = 28 V dc	f_{ab}	800		kHz
Subgroups 5 and 6						
Not applicable						

1/ For sampling plan, see MIL–PRF–19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3.
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) – for qualification and requalification only.

Inspection	MIL–STD–750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
End-point electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Steady-state operating life	1027	T_A = 25ºC, P_T = 0.6 W, V_{CE} = 30 V dc, t = 340 hours.	
End-point electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL–PRF–19500.	
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL–PRF–19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

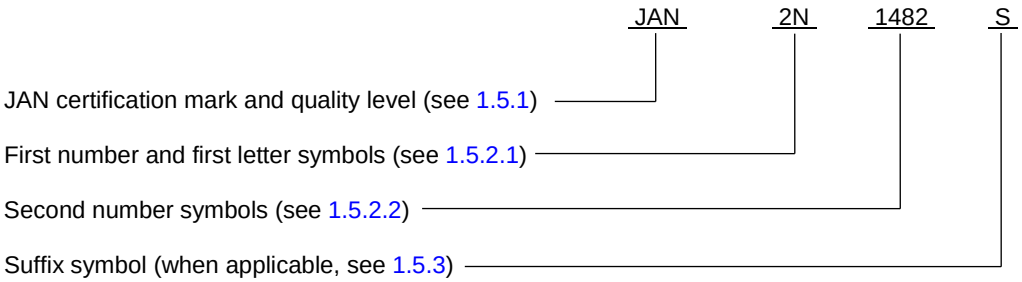
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for devices in a modified TO–205AA package (standard lead lengths)	PINs for devices in a modified TO–205AD package (short lead lengths)
JAN2N1479	JAN2N1479S
JAN2N1480	JAN2N1480S
JAN2N1481	JAN2N1481S
JAN2N1482	JAN2N1482S

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6.6 Supersession information and superseded PINs.

6.6.1 Lead finish. The original issue of this specification through MIL–S–19500/207C(EL), dated 9 December 1966, did not specify a lead finish.

6.6.2 Lead length. The original issue of this specification through MIL–S–19500/207C(EL), dated 9 December 1966, did not contain a suffix to designate lead length. This revision incorporates the "S" suffix option with the associated modified TO–205AD package. When applicable, PINs covering devices with a shortened lead length now shall include a suffix "S" to designate this package configuration (see 1.5.3 and figure 1.)

6.7 Application guidance.

a. The revised requirements in this document issue do not impair ready, bilateral replaceability between the transistor covered herein and the transistors, respectively, as covered in the previous issue(s) of this document.

b. To insure proper circuit application, particular attention should be given to the differential voltage-and-current requirements, ratings, and performance characteristics pertinent to the individual transistor types covered herein.

6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at "Semiconductor@.dla.mil" or by facsimile (614) 693–1642 or DSN 850–6939.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
Navy – EC
Air Force – 99
DLA – CC

Review activities:
Army – AR, MI, SM

Preparing activity:
DLA – CC

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